Controllable Real-Time Facial Expression Pipeline for 3D Avatars

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Abstract

Realistic emotional expression in 3D avatars is critical for virtual communication and human–computer interaction, yet most systems still rely on static, manually tuned facial controls. This paper presents a real-time pipeline that links facial Action Unit (AU) detection from live video with automated Valence–Arousal (V–A) estimation to drive a 3D avatar dynamically. The system uses OpenGraphAU to extract 41 AUs, maps them to 18 OpenFACS-compatible AUs through a calibrated bridge with baseline subtraction and temporal smoothing, and then applies V–A based emotion processing to support both emotion mirroring and controlled emotion inversion. Emotion processing operates in continuous V–A space using hand-crafted, FACS-informed rules with AU smoothing to improve temporal stability. Three modes are implemented: a normal mode that mirrors the user's emotion, an inverse-mirror mode that flips valence while preserving arousal, and an inverse-rotate mode that inverts both valence and arousal. The processed AU signals are transmitted over UDP to an Unreal Engine avatar at interactive frame rates, enabling webcam-driven facial animation on commodity hardware.

CCS Concepts

• **Computer systems organization** → *Real-time system architecture*; • **Human-centered computing** → **Visualization techniques**; **Systems and tools for interaction design**; **Interaction design process and methods**; **Interactive systems and tools**.

Keywords

Facial Action Units, real-time facial animation, avatar control, emotion inversion, affective computing

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1 Introduction

Virtual avatars have become increasingly prevalent in digital communication, virtual reality (VR), social platforms, and human-computer interaction applications. The COVID-19 pandemic accelerated the adoption of virtual communication technologies, revealing the critical importance of conveying emotional states in remote interactions. However, traditional avatar systems rely on static, manually controlled facial parameters or simplistic blendshape interpolation, lacking the dynamic emotional expressiveness necessary for authentic human connection.

The Facial Action Coding System (FACS) [8] provides a comprehensive framework for describing facial movements through Action Units (AUs). OpenFACS [5] and FACSvatar [27], are open-source implementations for avatar control, enabling AU-based facial animation. Current systems require users to manually adjust AU parameters through sliders or presets, making real-time emotion capture impractical. This limitation hinders applications requiring spontaneous emotional expression, such as virtual therapy sessions, remote education, empathy training, and social VR environments. Furthermore, emotion manipulation, particularly emotion inversion, remains an unexplored frontier in affective computing research. While systems for emotion recognition and synthesis exist independently, integrated frameworks that can detect, transform, and render emotions in real-time are rare. Emotion inversion has significant potential for psychological experiments investigating emotional incongruence, training scenarios for healthcare professionals managing difficult patient emotions, and entertainment applications requiring expressive character control.

Existing research in facial animation and avatar control exhibits several critical limitations that motivate this work. OpenFACS and similar systems [5, 27] provide robust AU-based control but require manual parameter adjustment. Real-time facial expression capture systems [4] exist but often use proprietary pipelines incompatible with open avatar platforms. The gap between AU detection and avatar control remains largely unbridged in accessible, open-source implementations. While emotion recognition has been extensively studied [15, 16], emotion transformation, especially inversion, lacks systematic investigation. Previous work on emotional mirroring in HCI [3] explored amplification and dampening but did not investigate complete polarity reversal. The psychological effects of sustained emotional incongruence in virtual communication remain poorly understood, limiting our ability to design effective training systems for emotional regulation. Most avatar expression systems lack comprehensive validation frameworks. Subjective user studies dominate evaluation methodologies [11], but objective, quantitative assessment using large-scale emotion datasets is uncommon. The disconnect between algorithm-level AU detection performance and render-level perceptual quality has not been systematically characterised.

This thesis makes several key contributions to affective computing and avatar animation research. We present a pipeline combining OpenGraphAU, custom AU processing, and OpenFACS for accessible, real-time emotion-driven avatar animation. We introduce a novel emotion inversion framework using V-A space manipulation that enables controlled emotion transformation with three operational modes, supported by calibrated AU-to-V-A and V-A-to-AU conversion functions. We establish a comprehensive dual-level validation methodology distinguishing algorithm performance in terms of AU detection and processing accuracy from system performance in terms of rendering quality and perceptual fidelity, validated on 1,287 emotion videos. We provide a comparative analysis of three smoothing methods, including EMA, Gaussian, and Geometric Mean, along with dynamic processing strategies such as baseline subtraction, emotion templates, and AU post-processing, with quantitative performance metrics. Finally, we deliver a working prototype for emotion incongruence experiments, empathy training systems, and communication psychology studies, with documented use cases and configuration guidelines.

Use cases include empathy training, where healthcare professionals practice responding to inverted patient emotions, entertainment where game NPCs react with opposite emotions for comedic effect, and research, where emotion perception and recognition are studied with controlled stimuli.

2 Related Works

2.1 Facial Action Unit Detection

The Facial Action Coding System (FACS) [8] established the foundation for systematic facial expression analysis by decomposing expressions into 44 anatomically-based Action Units (AU). The advent of deep learning revolutionised AU detection, with Convolutional Neural Networks (CNNs) enabling end-to-end learning from raw pixels [12]. Notable architectures include JAA-Net [24], which performs joint AU detection and intensity estimation using multi-scale feature aggregation and achieves state-of-the-art performance on BP4D [31] and DISFA [21] datasets. OpenGraphAU [19] employs graph-based relational reasoning over facial regions with a Swin Transformer backbone, modelling AU co-occurrence patterns explicitly and achieving F1-scores exceeding 0.65 on BP4D for most AUs. FAb-Net [28] implements a framework combining AU detection with facial landmark localisation for improved spatial consistency. While these models achieve high accuracy on benchmark datasets with AU annotations, applying them to real-time avatar control requires additional components for temporal smoothing, AU mapping, and protocol integration, which are gaps our system addresses.

2.2 Emotion Recognition Systems

Deep learning models for categorical emotion recognition, including happy, sad, and angry, have proliferated. The Affective Behaviour Analysis in-the-Wild (ABAW) challenges [15, 16] benchmark models on large-scale, diverse datasets, with top-performing systems employing Transformers (ViT) [18] and ensemble learning [30]. Valence-Arousal (V-A) space provides continuous emotion representation [23]. EmoNet [26] predicts continuous V-A coordinates alongside discrete emotion probabilities using ResNet architectures. AffectNet [22], the largest annotated emotion dataset with 450K images, enables training robust V-A regressors. Unlike end-to-end emotion classifiers, our system explicitly bridges AU detection and V-A estimation through custom mapping functions, enabling controllable emotion transformation impossible with black-box models.

2.3 Avatar Animation and Control

Earlier avatar systems used keypoint-based facial tracking, such as facial landmarks to fit blendshape parameters [6, 10]. While relatively fast, these methods lack semantic meaning and require careful manual tuning. Performance capture systems like Faceshift and Digital Emily [1] achieve photorealistic facial animation but require specialised hardware, including depth cameras and marker-based tracking, making them prohibitively expensive for general use. AU-based control provides anatomically meaningful and cross-compatible parameters. OpenFACS [5] and FACSvatar [27] implements FACS-compliant avatar animation. However, both methods focus on manual AU control via GUI sliders, lacking integration with automatic AU detection systems. LiveLink Face (Apple ARKit) [2] provides real-time facial tracking for iOS devices, transmitting 52 blendshapes to compatible avatars. Similarly, Metahuman Creator [9] supports expression transfer to avatars through blendshapes. While high-quality, this approach is platform-locked and does not expose AU-level control. No open-source system integrates state-of-the-art AU detection with OpenFACS for real-time, cross-platform avatar animation. Our work bridges this gap with a modular pipeline supporting CPU/GPU execution and extensible emotion processing.

2.4 Emotion Transformation and Manipulation

Transformed nonverbal behaviours in collaborative virtual environments (CVE) were introduced by [3]. Their system amplified or dampened detected emotions but did not invert polarity. Results showed that subtle expression manipulation improved negotiation outcomes without being consciously detected by users. Face2Face [25] enables real-time facial reenactment, transferring expressions from source to target identities. While visually impressive, the system operates at the pixel level without semantic AU representation, limiting controllability for psychological experiments. SyleGAN [14] based methods [13, 29] learn how to transfer expressions between a target and a driver video. However, the lack of real-time capability and interpretability restricts research applications. We introduce emotion inversion operating in V-A space, enabling controlled polarity reversal while maintaining AU-level interpretability. Three inversion modes, including normal, inverse-mirror, and inverse-rotate, support diverse experimental designs, with real-time performance suitable for interactive psychology studies.

2.5 Validation Methodologies

Most avatar expression systems rely on subjective user studies evaluating perceptual quality, naturalness, and emotional expressiveness [11]. While ecologically valid, these methods are labour-intensive, sample-limited, and suffer from inter-rater variability. Quantitative evaluation typically uses standard metrics such as F1-score and accuracy on annotated AU datasets, including BP4D [31], DISFA [21], and EmotioNet [21], emotion metrics including categorical accuracy and confusion matrices on FER-2013 [7] and AffectNet [22], and reconstruction metrics including PSNR and SSIM for image-based reenactment systems [25]. Existing metrics focus on recognition or reconstruction accuracy but do not assess the complete detection-to-rendering pipeline. Our dual-level validation framework with algorithm-level AU/V-A correlation and render-level avatar fidelity addresses this gap, quantifying degradation introduced by rendering and re-detection.

2.6 Datasets for Emotion and AU Analysis

AU-annotated datasets include BP4D [31] with 41 subjects and 328 videos featuring frame-level AU intensity annotations, though laboratory-controlled with limited diversity. DISFA [21] contains 27 subjects watching emotion-eliciting videos with continuous AU intensity labels, providing high-quality annotations but small scale. FEAFA [19] represents a large-scale AU dataset with 125K images used to train OpenGraphAU, covering 27 main AUs and 14 lateralized AUs. Emotion-annotated datasets include AffectNet [22] with 450K images featuring discrete emotions and V-A annotations under in-the-wild conditions, RAF-DB [17] with 30K facial images with crowdsourced emotion labels in real-world scenarios, and ENTERFACE [20] with 1,287 videos representing 44 subjects across 6 emotions with multiple sentences, designed for multimodal emotion recognition but lacking AU annotations while providing emotion ground truth.

3 Methods

Our approach integrates state-of-the-art AU detection with custom emotion processing and established avatar control protocols. The pipeline consists of five core modules operating in sequence. The Face Detection Module uses the OpenCV Haar Cascade detector to identify and track the largest face in the video stream. The AU Detection Module employs OpenGraphAU with Swin Transformer backbone to predict 41-dimensional AU intensity vectors from detected face regions. The AU Mapping Module implements a custom mapping function that converts from 41 source AUs to 18 target AUs, synthesising missing units such as AU28 via heuristics and AU45 via the blink scheduler. The Emotion Processing Module estimates V-A coordinates from the 18-dimensional AU vector, optionally applies emotion inversion, selects calibrated emotion templates, and post-processes AUs for enhanced expressiveness. The Avatar Control Module scales AUs and transmits control signals via UDP/JSON to an OpenFACS-compatible 3D avatar in Unreal Engine.

3.1 Emotion Inversion Design

The emotion transformation operates in Valence-Arousal space through the function $(V, A) = f_{\text{VA}}(\mathbf{a}_{18})$ where f_{VA} estimates continuous V-A coordinates from the 18-dimensional AU vector using calibrated weights. Inversion modes apply different transformations: Normal mode preserves the original emotion with $(V', A') = (V, A)$ representing identity mapping, Inverse-Mirror mode flips valence while preserving arousal with $(V', A') = (-V, A)$, and Inverse-Rotate mode performs 180-degree rotation with $(V', A') = (-V, -A)$. Inverted AUs are synthesized via $\mathbf{a}'_{18} = g_{\text{AU}}(V', A')$ and blended with original AUs at configurable strength $\lambda \in [0, 1]$ with $\mathbf{a}_{\text{out}} = (1 - \lambda)\mathbf{a}_{18} + \lambda \mathbf{a}'_{18}$.

The system must map from 41 source AUs to 18 target AUs. Let $\mathbf{a}_s \in \mathbb{R}^{41}$ be the source AU vector and $\mathbf{a}_t \in \mathbb{R}^{18}$ be the target AU vector, so we have to define a mapping function $g : \mathbb{R}^{41} \rightarrow \mathbb{R}^{18}$ that uses the estimated AU values to drive the avatar.

Most AUs have direct correspondence. AU28 (Lip Suck) and AU45 (Blink) require special handling since they are not predicted by OpenGraphAU. AU28 is approximated as AU28 = $\max(0, \text{AU23} - 0.5 \times \text{AU25})$. AU45, representing blinking, is synthetically generated. The blink patterns are generated with inter-blink interval $T_{\text{interval}} \sim \mathcal{U}(3.0, 6.0)$ seconds, blink duration $T_{\text{blink}} = 0.18$ seconds, and blink intensity $I_{\text{blink}} = 1.5$ during blink. This approximates the natural human blink frequency of 12-20 blinks per minute.

3.2 Signal Processing and Smoothing

Four signal processing techniques are utilised to improve the animation quality:

Baseline Subtraction. During the initial warmup period of N frames (default $N = 10$), the system computes a baseline:

$$\mathbf{b} = \frac{1}{N} \sum_{t=1}^{N} \mathbf{a}_t \tag{1}$$

Subsequent AU values are adjusted: $\mathbf{a}'_t = \max(\mathbf{0}, \mathbf{a}_t - \mathbf{b})$ This removes the individual's neutral expression bias, ensuring only expressive movements are captured.

The AU detection model prediction has a slight variance in the estimated values, which would translate to jerky motion on the avatar. To reduce this temporal jitter, the system implements three smoothing methods:

Exponential Moving Average (EMA).

$$\mathbf{a}_t^{\text{smooth}} = \alpha \mathbf{a}_t + (1 - \alpha)\mathbf{a}_{t-1}^{\text{smooth}} \tag{2}$$

where $\alpha \in [0, 1]$ is the smoothing factor (default $\alpha = 0.3$). Only requires storing one previous state ($O(1)$ memory).

Gaussian Filter.

$$\mathbf{a}_t^{\text{smooth}} = \sum_{i=0}^{W-1} w_i \mathbf{a}_{t-i} \tag{3}$$

where w_i are Gaussian weights: $w_i = \frac{1}{Z} \exp\left(-\frac{(i-\mu)^2}{2\sigma^2}\right)$, $\sigma = W/3$, W is window size (default $W = 5$), and Z is the normalization constant. The smoothing also minimises possible overdrive of the avatar's control parameters.

Figure 1: Demonstration of the three emotion inversion modes for happiness. No transformation (left), inverse-mirror where the emotion is flipped along the valence (positive-negative) axis (middle), to keep the energy of the expression, and 180° rotation in the valence-arousal space (right).

Geometric Mean Filter.

$$\mathbf{a}_t^{\text{smooth}} = \exp\left(\frac{1}{W}\sum_{i=0}^{W-1}\log(\mathbf{a}_{t-i}+\epsilon)\right)-\epsilon \tag{4}$$

where $\epsilon = 10^{-6}$ ensures numerical stability. This method is robust to outliers and reduces flickering in noisy camera conditions.

4 System Demonstration

The project provides a Tkinter-based configuration panel on startup. The GUI exposes the main runtime parameters: emotion mirroring mode (normal, inverse mirror, inverse rotate, see Figure 1), camera index, GPU ID, scaling factor, smoothing method and window size, baseline length, and UDP target (IP and port). It can launch and stop the avatar bridge, optionally show the camera preview, and log Valence-Arousal and AU values for debugging. User preferences (e.g. last-used mode and smoothing settings) are stored in a small JSON file under the home directory, so typical settings persist between sessions. This interface is intended for everyday use and live demos, while the CLI remains available for scripted experiments.

When the application launches, OpenFACS opens in full-screen mode and starts to mirror the user's interaction based on the selected mirroring mode.

The application uses the built-in camera of the device, and the video stream goes through multiple pre-processing steps before it reaches the avatar. The pipeline samples frames from the stream as the same rate as the FPS, detects the available faces on the image and selects the face with the largest area (assuming that the user sit the closest to the camera). The selected face is cropped from the frame and goes to an AU estimator. Then, the AUs are accumulated depending on the selected smoothing method and transformed based on the user-selected mirroring mode. To enhance the expressiveness of the avatar, the smoothed-transformed values go through a clipping and scaling, then broadcast to the avatar.

5 Implementation Details

The main goal of the avatar is to be interactive in real-time, on commodity hardware. We have tested the pipeline on multiple machines and measured the approximate time cost for each of the sub-tasks. By this, the end-to-end latency can be decomposed into the following stages stages:

$$T_{\text{total}} = T_{\text{capture}} + T_{\text{detect}} + T_{\text{au}} + T_{\text{process}} + T_{\text{transmit}} \tag{5}$$

where $T_{capture}$ is the camera capture (≈ 5-10 ms), T_{detect} is the face detection (≈ 5-8 ms), T_{au} is the AU inference (≈ 30-50 ms), $T_{process}$ is the AU mapping and smoothing in time (≈ 1-2 ms), and $T_{transmit}$ is the UDP transmission latency/time. Generally, the AU inference (T_{au}) dominates the time spent in the pipeline. The UDP transmission time can also vary depending on where we broadcast the AU values. In our case, we used the local broadcasting (the avatar is on the same physical computer as the pipeline), but it is also possible to send the predicted values to a remote machine, where the avatar is running. A typical Latency Breakdown for the pipeline (CPU mode, local broadcasting):

Table 1: Performance benchmarks shown on different hardware. We have tested the pipeline on commodity hardware. The target FPS has two interpretations here. The first is how often we update the avatar's expression. However, these updates rely on the AU detection, which takes the most time in our pipeline. Of course, the avatar can be updated faster, but since the AU detector wouldn't finish the prediction in time, the avatar would be updated with the old values. Similarly, the latency is mostly dependent on the hardware, since the prediction model works faster on a GPU.

Hardware	Avg. FPS	Avg. Latency (ms)	P95 Latency (ms)
Intel i5-8250U (CPU)	18.3	54.6	68.2
Intel i7-9750H (CPU)	24.1	41.5	52.3
NVIDIA GTX 1060 (GPU)	35.7	28.0	34.5
NVIDIA RTX 3070 (GPU)	52.3	19.1	23.8

The system was also tested on a 1,800-frame test to see that the system is independently validated at a higher emotional semantic level. This ensured, that the system's control is individual independent.

6 Limitations and Future Works

Our method comes with a few limitations. Haar Cascade struggles with profile views and extreme lighting, AU28 approximation where heuristic-based approximation may not capture true lip suck, blink timing where synthetic blinks do not synchronize with actual eye closures, lateralized AUs where the current implementation does not preserve left and right AU asymmetry, occlusion handling where partial occlusions such as hand on face may produce artifacts, and multi-subject tracking where there is no support for tracking multiple faces simultaneously. The system achieves reliable real-time facial animation for single-user scenarios.

7 Acknowledgements

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